

ST PAUL AGCM, AB, Canada

WMO: 712750

Lat: 54.0111N

Lon: 111.2719W

Elev: 649

StdP: 93.77

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-33.6	-30.2	-37.1	0.1	-33.5	-33.5	0.2	-30.1	9.5	-7.5	8.6	-8.2	1.4	230	0.687

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	27.5	17.4	25.7	16.7	24.0	16.1	19.2	25.3	18.0	23.7	16.9	22.3	3.9	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.8	13.0	22.1	15.7	12.1	20.7	14.7	11.3	19.3	57.3	25.2	53.3	23.7	49.9	22.4	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	8.1	7.2	DB	-38.9	30.3	2.9	1.3	-41.0	31.2	-42.7	31.9	-44.4	32.6	-46.5	33.6	(4)
(5)			WB	-38.6	21.7	2.9	1.0	-40.7	22.4	-42.4	23.0	-44.0	23.5	-46.1	24.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.6	-13.6	-13.8	-6.5	2.4	10.1	14.3	16.7	15.1	10.2	3.3	-6.3	-13.3	(6)	
	DBStd	12.77	8.72	7.72	7.41	5.44	4.46	3.13	2.83	3.28	4.46	4.78	7.00	8.03	(7)	
	HDD10.0	3682	733	666	511	234	54	4	0	3	51	213	489	723	(8)	
	HDD18.3	6131	991	899	769	479	256	126	68	109	246	467	739	981	(9)	
	CDD10.0	623	0	0	0	5	57	133	207	159	56	5	0	0	(10)	
	CDD18.3	31	0	0	0	0	1	4	16	8	1	0	0	0	(11)	
	CDH23.3	502	0	0	0	1	38	72	217	123	49	2	0	0	(12)	
	CDH26.7	85	0	0	0	0	4	8	46	17	9	0	0	0	(13)	
(14) Wind	WSAvg	3.3	3.2	3.0	3.4	3.8	3.8	3.4	3.1	3.0	3.2	3.4	3.3	3.0	(14)	
(15) Precipitation	PrecAvg	437	19	14	18	31	43	76	83	62	36	22	20	19	(15)	
	PrecMax	526	50	32	35	86	81	162	183	138	92	56	56	36	(16)	
	PrecMin	278	0	1	5	2	4	22	32	22	3	5	2	1	(17)	
	PrecStd	67	11	7	9	17	21	33	28	26	22	11	14	9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.0	4.9	10.3	21.2	27.1	27.8	30.2	29.0	28.4	22.1	11.1	4.9	(19)	
		MCWB	2.3	1.5	5.1	10.5	14.0	16.5	19.4	18.6	16.0	12.1	5.6	2.0	(20)	
	2%	DB	2.8	2.1	6.6	17.1	24.0	25.4	27.8	26.2	24.3	16.4	6.7	2.2	(21)	
		MCWB	0.7	-0.1	3.1	8.4	12.8	15.8	18.7	17.5	14.9	9.4	3.5	0.1	(22)	
	5%	DB	0.9	-0.2	4.7	13.8	21.4	23.3	25.8	24.5	21.4	13.4	4.0	0.0	(23)	
		MCWB	-0.6	-1.6	1.7	6.7	11.5	14.7	17.8	17.1	13.7	7.9	1.5	-1.6	(24)	
	10%	DB	-1.4	-2.4	2.8	11.1	19.3	21.4	23.8	22.7	18.5	10.9	2.0	-2.4	(25)	
		MCWB	-2.6	-3.7	0.5	5.3	10.4	13.8	17.0	16.1	12.1	6.3	0.3	-3.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.4	2.0	5.6	11.1	15.1	18.5	21.2	20.5	16.9	12.6	6.4	2.4	(27)	
		MCDB	4.6	4.4	9.6	19.9	23.7	24.5	26.9	25.8	25.7	20.9	10.6	4.8	(28)	
	2%	WB	1.0	0.1	3.1	8.9	13.4	17.0	19.8	18.7	15.5	10.2	3.9	0.3	(29)	
		MCDB	2.5	1.7	6.2	16.0	21.5	22.7	26.1	24.5	22.8	15.5	6.6	2.2	(30)	
	5%	WB	-0.5	-1.7	1.8	7.1	12.3	15.9	18.6	17.6	14.1	8.2	1.8	-1.5	(31)	
		MCDB	0.9	-0.1	4.2	13.3	19.9	21.4	24.5	23.1	20.5	12.3	3.6	-0.1	(32)	
	10%	WB	-2.6	-3.8	0.8	5.6	11.1	14.8	17.4	16.5	12.7	6.9	0.3	-3.6	(33)	
		MCDB	-1.3	-2.6	2.8	10.6	17.6	20.0	22.7	21.6	17.7	10.6	1.8	-2.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.9	11.0	10.3	10.7	12.8	11.3	11.9	12.2	12.2	10.5	8.4	9.0	(35)	
		MCDBR	10.4	12.6	11.8	15.4	16.1	14.6	15.1	15.3	17.3	15.3	10.5	11.1	(36)	
	5% WB	MCWBR	8.8	10.8	8.6	8.4	7.3	7.1	7.7	7.9	8.8	9.1	7.5	9.5	(37)	
		MCDWR	10.4	12.8	10.8	14.6	14.5	12.4	13.5	13.5	15.8	14.4	10.0	11.2	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.250	0.270	0.289	0.327	0.345	0.361	0.360	0.369	0.308	0.275	0.249	0.223	(40)	
		taud	2.295	2.279	2.348	2.375	2.377	2.351	2.392	2.330	2.502	2.565	2.448	2.331	(41)	
	Ebn at Noon		732	834	899	904	901	887	881	848	873	839	747	706	(42)	
		Edh at Noon	56	80	94	106	112	116	109	110	81	60	48	43	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.91	1.96	3.48	4.67	5.75	5.88	6.01	4.90	3.44	1.95	0.92	0.66	(44)	
	RadStd		0.10	0.15	0.26	0.40	0.45	0.42	0.33	0.38	0.37	0.30	0.11	0.06	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (59 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page